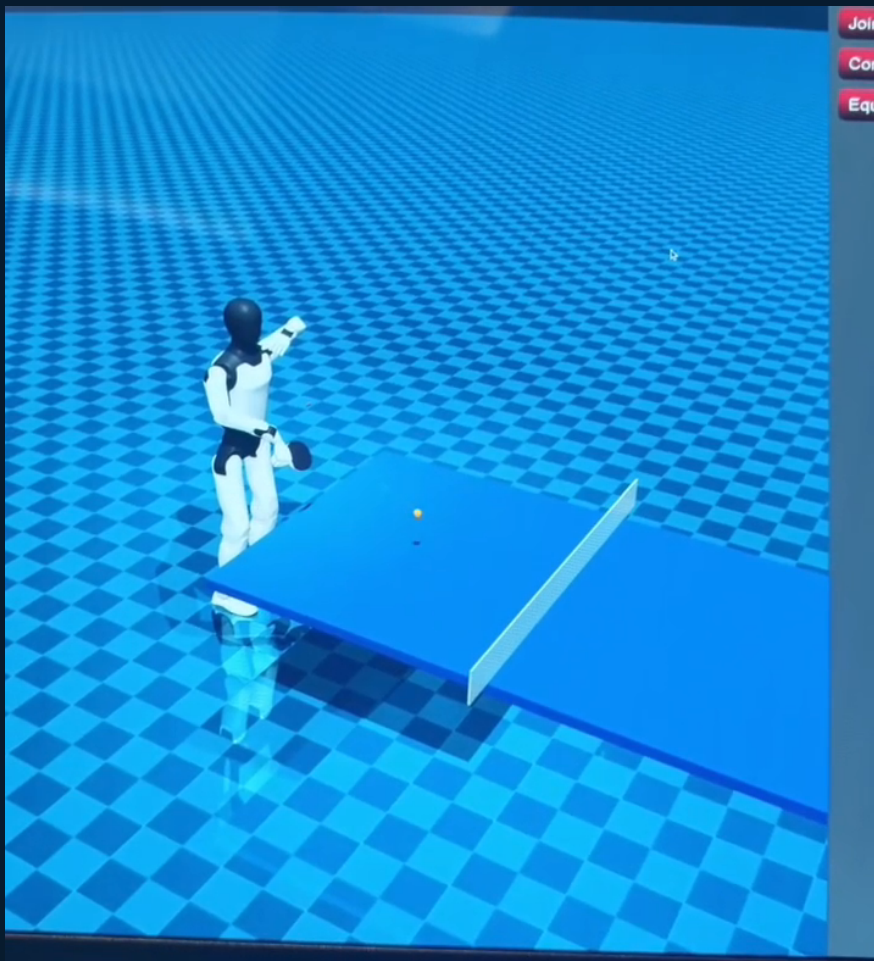


HOPE / OPEN-SOURCE HUMANOID PING-PONG

A3 MuJoCo Serve Package

High-level build, verification, and on-robot execution guide

PR #18 reference application: fully tested on Agibot A3 / executable / safe



Visual reference captured from the supplied 6.525-second PR #18 demo. Every build and run step in this guide uses the exact machine-validated PR #18 motion.

HOPE nightly_built / 31 July 2026

01 / PACKAGE PURPOSE

One validated motion, from package to A3 execution

This operator walkthrough treats the original PR #18 CSV as immutable and follows it through identity verification, native-MDU compilation, guarded preflight, and execution.

1. Identify Select the exact original PR #18 CSV under assets/validated/.	2. Verify Match its Git blob and SHA-256 to the machine-tested source.	3. Compile Build the high-level arm runner natively on the A3 MDU.
4. Preflight Verify package hashes, AArch64 runtime, limits, topics, and vendor state.	5. Prepare Check hold behavior and move both arms to the committed READY pose.	6. Serve Trigger the original fixed motion and restore vendor ownership afterward.

Important ownership boundary

The CSV contains every SDK joint column, but the validated high-level runtime publishes only the 14 arm joints. The installed A3 vendor stack retains control of the waist, legs, and neck.

Path	Purpose
assets/validated/	Exact PR #18 CSV, its runtime contract, identity notes, and demo reference.
runtime/	Native-MDU C++ runner, protected wrapper, and package builder.
runtime/README.md	Detailed preflight, operator modes, handoff, and restoration behavior.
tests/	CSV identity, manifest provenance, limits, and runtime-wrapper regression checks.
README.md	Package orientation and technical reference links.

Application root: apps/a3_mujoco_serve/

02 / IMMUTABLE ASSET IDENTITY

Confirm the exact PR #18 motion

The relocated file is not a modified approximation. It is the same byte sequence as the machine-tested source and remains the default input to the package builder.

1

Byte-for-byte preserved

assets/validated/serve_policy.csv is byte-for-byte identical to the original PR #18 CSV. Its SHA-256 is 2a7de3f1c97a300069899c139c9eb96e94fd61d3419701d5e44ef37b2bf6641d. The exact application is fully tested, executable, and safe on Agibot A3.

2

Git object identity also matches

The original PR #18 path and the reorganized validated path both resolve to Git blob 200a52552feb3ed5d978310fb2d95ca6150aff2e. This proves the relocation did not change headers, values, formatting, line endings, or represented joint motion.

3

Manifest-bound default

The validated manifest records the same SHA-256. The build script selects this CSV and this manifest by default, then checks the file identity before compiling or packaging anything.

Validated build command

Run the builder with its default validated asset selection:

```
bash runtime/scripts/build_a3_app.sh --jobs 2
```

Evidence	Validated reference
Motion identity	Original and relocated files share the same Git blob and SHA-256.
Transport	PR #18 high-level /motion/control/arm_joint_command runtime.
Video reference	pr18_a3_serve_demo.mp4, SHA-256 5e476beb...5d36ac8.
Regression tests	24 tests pass, including asset hashes and runtime-wrapper behavior.

03 / FAST PATH

Rebuild the validated example on the A3 MDU

Perform these commands on the robot's ARM64/aarch64 MDU. The default builder input is the exact PR #18 CSV documented on the preceding page.

1

Open the application on the MDU

Use a complete HOPE checkout containing apps/a3_mujoco_serve. Build natively in the installed vendor ROS 2 environment; this is not a macOS or generic desktop build.

```
cd /path/to/HOPE/apps/a3_mujoco_serve
source /agibot/software/v0/entry/env/env.sh
```

2

Build the default validated asset

The builder verifies that the CSV hash matches the selected manifest, compiles the C++ runner, and creates a manifest-bound deployment directory.

```
bash runtime/scripts/build_a3_app.sh --jobs 2
```

3

Enter the isolated package

The output contains the AArch64 runner, wrapper, common checks, exact CSV, motion manifest, build identity, and a SHA-256 package manifest.

```
cd dist/a3_serve_vendor_arm
ls -la
```

4

Run read-only preflight first

The default command validates package hashes, executable architecture and libraries, trajectory limits, vendor action, topics, and motion_player state. It does not publish robot commands.

```
./run_a3_app.sh
```

Stop if preflight does not pass

Do not add the real-command confirmation flag to work around an error. Correct the package, vendor environment, robot action, topic ownership, or motion_player state first.

04 / ON-MACHINE EXECUTION

Run the example motion in guarded stages

Real modes require an interactive terminal and the literal `--confirm-real-commands` argument. Use a trained operator, working e-stop, physical support, and a cleared workspace.

Stage	Command	Operator outcome
Preflight	<code>./run_a3_app.sh</code>	Read-only verification. No arm-command publisher is created.
Hold	<code>./run_a3_app.sh --hold-only --confirm-real-commands</code>	Checks a short measured-state hold.
Prepare	<code>./run_a3_app.sh --prepare-only --confirm-real-commands</code>	Moves both arms to READY and holds until Ctrl-C.
Serve	<code>./run_a3_app.sh --serve-only --confirm-real-commands</code>	Moves to READY, waits for Space, and plays the validated serve/return.

Recommended sequence

```
./run_a3_app.sh
./run_a3_app.sh --hold-only --confirm-real-commands
./run_a3_app.sh --prepare-only --confirm-real-commands
./run_a3_app.sh --serve-only --confirm-real-commands
```

READY	TRIGGER	STRIKE	RESTORE
The runner prepares both arms using the committed reference pose.	In serve-only mode, press Space at physical ball release.	The runner holds READY for 1.000 s; nominal strike is about 1.060 s after Space.	On exit or failure, the wrapper stops its publisher before restoring motion_player.

What the wrapper protects

It prearms the runner, stops only motion_player through the vendor process manager, verifies publisher handoff, releases the custom publisher, and restores motion_player after the custom publisher is stopped. The left wrist-roll remains at its measured entry value.

Before every real run

Confirm e-stop reachability / safety rope or support / clear arm and racket envelope / correct ball-release operator / no unexpected command publisher / robot remains in MotionControlAction_MOTION/RUNNING.

05 / PREFLIGHT CONTRACT

What the package checks before motion

The default invocation is intentionally read-only. It verifies that the original validated asset, compiled runner, vendor environment, and ROS ownership state agree before any real mode is allowed.

Package identity Verify every packaged file against the SHA-256 package manifest.	Motion identity Verify the exact CSV against source.sha256 in the validated manifest.	Executable Require an AArch64 ELF with no unresolved shared libraries.
Trajectory Validate finite values, A3 high-level limits, frame count, steps, and speed gates.	Vendor state Require the expected A3 model, motion action, and motion_player IDLE state.	ROS ownership Check state and command topics before the guarded publisher handoff.

Expected isolated package

```
dist/a3_serve_vendor_arm/
  a3_serve_vendor_arm_runner
  run_a3_app.sh
  a3_app_common.sh
  motions/serve_policy.csv
  config/serve_vendor_arm_manifest.json
  config/serve_vendor_arm_build.env
  config/serve_vendor_arm_package.sha256
```

The preflight command is safe to repeat

Run `./run_a3_app.sh` after a rebuild, package copy, vendor-stack restart, or robot reboot. A passing preflight does not move the arms and does not replace `motion_player`'s publisher.

Failure behavior

A missing file, hash mismatch, wrong architecture, unresolved dependency, invalid trajectory, unexpected robot state, or publisher conflict stops the wrapper before real commands.

06 / VERIFICATION AND REFERENCES

What was checked in this local build

The local package was checked against the original PR #18 asset identity, validated manifest, runtime contracts, shell syntax, and portable wrapper tests.

Check	Observed result
Git blob	Original PR #18 path and relocated asset both use 200a52552feb3ed5d978310fb2d95ca6150aff2e.
SHA-256	2a7de3f1c97a300069899c139c9eb96e94fd61d3419701d5e44ef37b2bf6641d.
CSV contract	3,878 rows x 37 columns at 200 Hz; bytes and represented motion are unchanged.
Recorded limit metric	Validated manifest records 4.9879 rad/s command-stroke peak vs 5.2 rad/s source gate.
Runtime	Original PR #18 high-level arm runner and guarded handoff behavior retained.
Tests	24 passed, including asset hashes and runtime-wrapper behavior.

Primary technical references

Reference	Repository location
Application guide	apps/a3_mujoco_serve/README.md
A3 runtime guide	apps/a3_mujoco_serve/runtime/README.md
Validated assets	apps/a3_mujoco_serve/assets/validated/README.md
Original motion	apps/a3_mujoco_serve/assets/validated/serve_policy.csv
Validated manifest	apps/a3_mujoco_serve/assets/validated/serve_vendor_arm.json
Demo source	apps/a3_mujoco_serve/assets/validated/pr18_a3_serve_demo.mp4

Shortest validated-example reminder

```
cd /path/to/HOPE/apps/a3_mujoco_serve
source /agibot/software/v0/entry/env/env.sh
bash runtime/scripts/build_a3_app.sh --jobs 2
cd dist/a3_serve_vendor_arm
./run_a3_app.sh
./run_a3_app.sh --serve-only --confirm-real-commands
```

Use the full staged hold and prepare checks on first deployment or whenever the robot, vendor stack, package, workspace, or operator setup has changed.